EXPERIMENT-10

Aim : Demonstrate the working of temperature sensor - DHT11

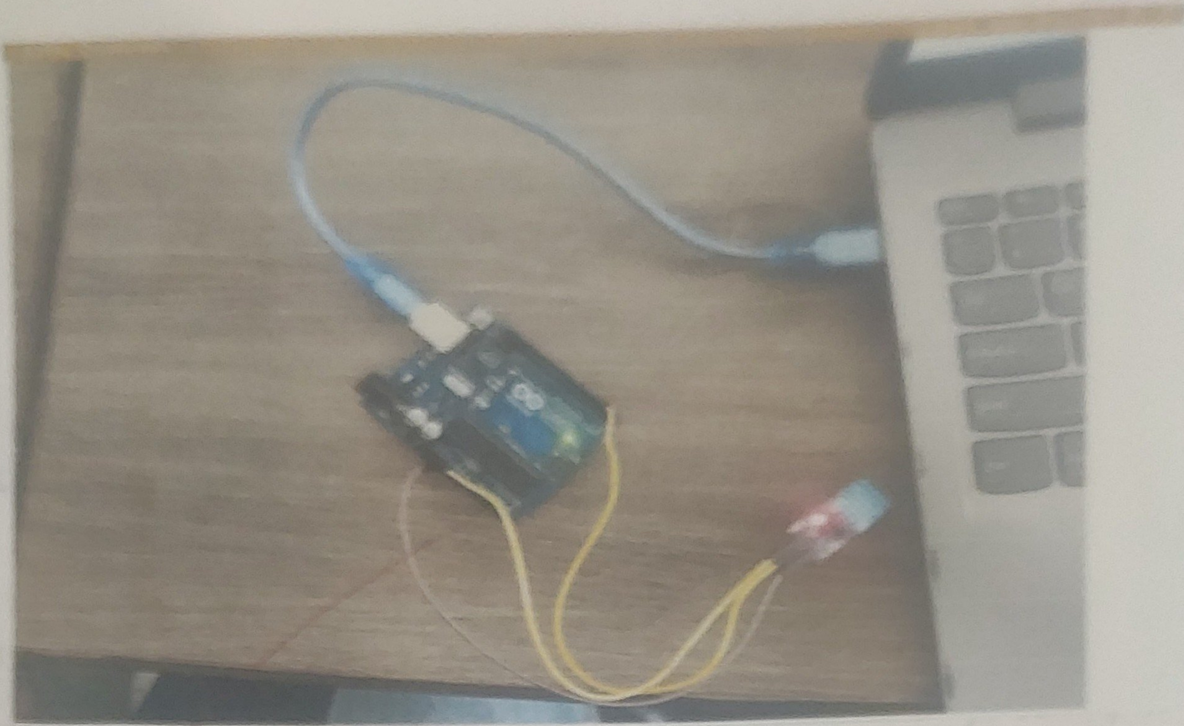
Appartus: Arduino UNO, DHT11, Temp, wires, USB cable.

Theory :

A temperature sensor is a piece of electronic equipment that detects the temp of its surroundings and transforms the incoming data into electronic output data to control record or signal temperature variations. There are several types of temprature sensor.

Some of them need direct constant with the physical target that is being identified, which is introduced as constact temperature sensor. In constract, others sense the temp of the target indirectly, which is described as non-contact temperature sensor.

Performance Result:



CODE :

```
# include < DHT 11.h>
DHT 11  dht11 (2) ;
void setup () {
    serial.begin (9600) ;
}
void loop () {
    int temperature = dht11.readTemperature();
    if (temperature != DHT 11 :: ERROR_ CHECKSUM
        && temperature != DHT11 :: Error_
        time OUT ) {
    serial.print ("Temperature: ") ;
    serial.print (temperature ) ;
    serial.println (" °C " ) ;
  } else {
    serial.println ( DHT11 :: get ErrorString
        (temperature )) ;
  }
}
```

conclusion:

The DHT 11 with Arduino UNO successfully measures and displays temperature, making it useful for basic environmental monitoring applications.

Used